

2.2 Applying Quotient Laws of Exponents to Algebraic Expressions

Lesson Introduction

	Benchmark	MA.912.NSO.1.2: Generate equivalent algebraic expressions using the properties of exponents.		Learning Target	I can generate equivalent algebraic expressions to find the quotient of two monomials using the properties (laws) of exponents.
	Benchmark Coherence	Building On MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency. MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.			Working Toward MA.912.NSO.1.3 Generate equivalent algebraic expressions involving radicals or rational exponents using the properties of exponents.
	Essential Questions	- What is the quotient of a power law of exponents and how do you apply it during the problem-solving process? - What is the power of a quotient law of exponents and how do you apply it during the problem-solving process? - Why is it important to consider the domain when generating equivalent algebraic expressions to find the quotient of two monomials?			
	Lesson Narrative	In this lesson, students will determine the equivalent expression that is the quotient of two monomials using the quotient laws of exponents. Students will analyze the quotient of a power law and power of a quotient law by first viewing the expression in expanded form prior to making the connection to the laws. During this lesson, students will be introduced to domain restrictions of an expression.			
	Component Summary	2.2.1 Warm-Up (~5 min.)	Explaining a “Would You Rather?” choice (<i>SE p. 62</i>)	2.2.4 Guided Instruction (5-10 min.)	Applying the power of a quotient law (<i>SE p. 65</i>)
		2.2.2 Guided Instruction (10 min.)	Applying the quotient of powers law (<i>SE p. 62</i>)	2.2.5 Activity (10-15 min.)	Matching expression to equivalent monomial expressions using the quotient exponent laws (<i>SE p. 66</i>)
		2.2.3 Guided Practice (5-10 min.)	Writing an equivalent expression and domain restrictions using the quotient of powers law (<i>SE p. 64</i>)	2.2.6 Your Turn! (10-15 min.)	Writing equivalent expressions using the quotient exponent laws (<i>SE p. 66</i>)
		2.2.7 Wrap-Up (~5 min.)	Reflection on the day’s lesson (<i>SE p. 67</i>)		
	Resources	Glossary		Materials	Additional Preparation
		<u>Key Terms:</u> - Quotient of Powers - Power of a Quotient <u>Additional Terms:</u> N/A		(Optional) Calculator (Optional) Colored Pencils	N/A

 Teacher Tips	Common Misconceptions - Students may not apply the power to the coefficient when applying the power of a quotient law. - Students may add the exponents when applying the power of a quotient law. - Students may not apply the negative exponent law.	Things to Consider - Understanding domain restriction is not the learning goal of the lesson, but it is important for future topics in mathematics. Students may need a reminder that you cannot divide by zero and why it is necessary to consider when applying the quotient exponent properties. - Consider encouraging students to try applying the laws and the expanded form to build deep conceptual understanding. - BEST uses “properties” and “laws” interchangeably. Both laws of exponents and properties of exponents need to be used during instruction to ensure students understand they are the same.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 2.2.4 Guided Instruction positions students to analyze other students’ work using the expanded form to discover the power of quotient law. Consider applying this routine on each set of student work, positioning students to discover the power of a quotient law.	MA.K12.MTR.5.1: Patterns and Structure Students will be positioned to look for patterns when learning to write equivalent expressions. In the 2.2.2 Guided Instruction, students will use the quotient of powers law and expanded form of the monomials to determine what is happening to the coefficients, exponents, and bases when two monomials are divided.
 Differentiate Instruction	Notice the opportunities within the 2.2.2 Guided Instruction for both verbal (partner discussion) and algebraic reasoning (questions 1a, 2a, and 3a). While some students might prefer the entry point of writing and making sense of the information internally, other students might prefer to process the information verbally before formalizing their observations.	
Lesson Notes:		

2.2.1 Warm-Up: Would You Rather?

Component Overview: Students engage in a “Would You Rather” activity between two scenarios. Display the prompt for the class and monitor as students answer the question independently prior to sharing their answer with their partner.



2.2.1 Warm-Up: Would You Rather?

1. Would you rather earn a salary of \$50,000 per year, while everyone else earns \$25,000, or earn a salary of \$100,000 per year, while everyone else earns \$200,000?



Answers will vary. I would like to make \$50,000 so that I am able to make more than others. I would like to make \$100,000 so that I can have more money.

2. Explain to your partner how you decided on the option you chose.

Answers will vary.



Provide students with the opportunity to independently read, process, and answer the prompt before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of the information: first for independent learning, then pairing for verbal processing in order to give opportunities to develop confidence, lastly for precise responses when students share out.

2.2.2 Guided Instruction: Quotient of Powers

Component Overview: Students are introduced to the quotient of powers law of exponents. Lead students through the Guided Instruction using both whole-group instruction and partner discussion. Utilize the built-in questioning and statements to position students to take the lead on most of the whole-group discussion.



2.2.2 Guided Instruction: Quotient of Powers

When dividing monomials, the **quotient of powers** law of exponents is used.

A breakdown of dividing 3^5 by 3^4 is shown.

$$\frac{3^5}{3^4} = \frac{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot 3}{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3}} = \frac{3}{1} = 3^1 = 3$$

A breakdown of dividing x^5 by x^2 is shown.

$$\frac{x^5}{x^2} = \frac{\cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x}{\cancel{x} \cdot \cancel{x}} = \frac{x \cdot x \cdot x}{1} = x^3$$

1. Consider the expression $\frac{x^2}{x^5}$.

- a. Complete the steps to find an equivalent expression.

$$\frac{x^2}{x^5} = \frac{\cancel{x} \cdot \cancel{x}}{\cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x} = \frac{1}{x^3}$$

- b. With your partner, discuss what value is in the numerator of the equivalent expression.

Answers will vary.



Quotient of Powers

$$\frac{a^m}{a^n} = a^{m-n} \text{ and conversely } a^{m-n} = \frac{a^m}{a^n}$$

- c. Use the quotient of powers law to find an equivalent expression for $\frac{x^2}{x^5}$. Show your work.

$$\begin{aligned} \frac{x^2}{x^5} \\ &= x^{2-5} \\ &= x^{-3} \\ &= \frac{1}{x^3} \end{aligned}$$



Consider utilizing the structure of the examples at the beginning of the lesson. Use think-aloud strategies to explain why certain variables and exponents have been crossed out while demonstrating the quotient of powers law.



Consider asking the following:

- For question 1a, why is the numerator 1 and not 0?
- When using the quotient of powers law, can you think of a circumstance when the numerator would be 0? Why or why not?

2.2.2 Guided Instruction: Quotient of Powers (continued)

- d. Discuss with your partner how the expression found using the quotient of powers and the expression found when rewriting $\frac{x^2}{x^5}$ as $\frac{x \cdot x}{x \cdot x \cdot x \cdot x \cdot x}$ are the same.

Answers will vary. The difference between the exponents when using the quotient of powers property ($2 - 5$) is -3 , which means there will be a variable with an exponent of 3 left in the denominator. In the expanded form, two variables can be canceled from the numerator with two of the same variables in the denominator leaving a product of 3 variables left in denominator.

2. Consider the expression $\frac{x^3 y^{11} z^3}{x^2 y^2}$.

- a. Use the property of exponents to find an equivalent expression.

$$\frac{x^3 y^{11} z^3}{x^2 y^2} = x^{(3-2)} y^{(11-2)} z^{(3)} = x^{\boxed{1}} y^{\boxed{9}} z^3$$

- b. Discuss with your partner why the power of z did not change.

The power of z did not change because there were no other variables of z to combine with.

3. Consider the expression $\frac{12x^3 y^1}{6x^1 y^3}$.

- a. Determine an expression equivalent to $\frac{12x^3 y^1}{6x^1 y^3}$.

$$\begin{aligned} \left(\frac{12}{6}\right) \left(\frac{x^3}{x^1}\right) \left(\frac{y^1}{y^3}\right) &= \\ \frac{2x^2 y^{-2}}{y^2} &= \end{aligned}$$



Consider challenging students to write their own expression that would utilize the quotient of powers law to simplify and also has domain restrictions.



Consider discussing if there is only one approach to dividing monomials. Stress that both methods (exponent laws and expanded form) can be used and that either pathway done correctly will lead to the correct simplified expression.

2.2.2 Guided Instruction: Quotient of Powers (continued)

- b. The variable y is in the denominator of the expression equivalent to $\frac{12x^3y^1}{6x^1y^3}$. Discuss with your partner what value(s), when substituted in for y , would make the expression undefined.

Answers will vary. If the value of 0 was substituted for the value of y , the expression would be undefined because you can't divide by 0.

Domain restrictions should be written to exclude the value(s) which would make the expression undefined. For example, in the expression $\frac{1}{x^3}$, when 0 is substituted for x the expression is undefined, making 0 a domain restriction. The restriction for $\frac{1}{x^3}$ is represented as $x \neq 0$.

- c. Write the domain restriction for the expression,

$$\frac{12x^3y^1}{6x^1y^3}$$

$$x \neq 0; y \neq 0$$

2.2.3 Guided Practice: Generating Equivalent Algebraic Expressions Using Quotient of Powers

Component Overview: Students work to generate equivalent algebraic monomial expressions using the quotient of powers property of exponents. Monitor students as they work and use questioning if students need help finding an entry point to a problem.



2.2.3 Guided Practice: Generating Equivalent Algebraic Expressions Using Quotient of Powers

- Complete the table for each expression.

Expression	Equivalent Expression	Domain Restriction?	If yes, state the restriction.
$\frac{12s^5t^9}{20t^3}$	$\frac{3s^5t^6}{5}$	☐ yes ☐ no	$t \neq 0$
$\frac{21a^8b^3c^{15}}{7a^3b^{12}c^2}$	$\frac{3a^5c^{13}}{b^9}$	☐ yes ☐ no	$a \neq 0; b \neq 0; c \neq 0$
$\frac{3nm^4}{6n^2m^7}$	$\frac{1}{2nm^3}$	☐ yes ☐ no	$n \neq 0; m \neq 0$



Students who struggle with problems that contain multiple variable factors may benefit from separating and organizing the factors prior to attempting to apply the exponent laws. For example, $\frac{12s^5t^9}{20t^3}$ may be separated into a table as shown.

coefficient	variable s	variable t
$\frac{12}{20}$	s^5	$\frac{t^9}{t^3}$
$\frac{3}{5}$	s^5	t^6
Answer: $\frac{3s^5t^6}{5}$		



Challenge students who finish early to choose one of these expressions to create an Error Analysis problem for others to analyze and solve. Creating Error Analysis problems positions students to think deeply about the content and the common mistakes and errors that occur.

2.2.4 Guided Instruction: Power of a Quotient

Component Overview: Students are introduced to the power of a quotient property of exponents. Lead students through the Guided Instruction using whole-group instruction. Apply a “You Do, We Do” approach where students read and attempt a question prior to holding a whole-group discussion for each question.



2.2.4 Guided Instruction: Power of a Quotient

- Three students generated equivalent expressions using exponent laws.
 - Finish Amir’s work using a similar strategy to the one used by Luis and Maya.

Luis	Maya	Amir
$\left(\frac{3}{11}\right)^5$ $\frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11}$ $\frac{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3}{11 \cdot 11 \cdot 11 \cdot 11 \cdot 11}$ $\frac{3^5}{11^5}$	$\left(\frac{x}{6y}\right)^2$ $\frac{x}{6y} \cdot \frac{x}{6y}$ $\frac{x \cdot x}{6y \cdot 6y}$ $\frac{x^2}{6 \cdot 6 \cdot y \cdot y}$ $\frac{x^2}{36y^2}$	$\left(\frac{7c}{r}\right)^3$ $\frac{7c}{r} \cdot \frac{7c}{r} \cdot \frac{7c}{r}$ $\frac{7c \cdot 7c \cdot 7c}{r \cdot r \cdot r}$ $\frac{7 \cdot 7 \cdot 7 \cdot c \cdot c \cdot c}{r \cdot r \cdot r}$ $\frac{343c^3}{r^3}$

- Complete the table.

Expression	Equivalent Expression	Domain Restriction?	If yes, state the restriction.
$\left(\frac{3}{11}\right)^5$	$\frac{3^5}{11^5}$	☐ yes ☐ no	N/A
$\left(\frac{x}{6y}\right)^2$	$\frac{x^2}{36y^2}$	☐ yes ☐ no	$y \neq 0$
$\left(\frac{7c}{r}\right)^3$	$\frac{343c^3}{r^3}$	☐ yes ☐ no	$r \neq 0$

- Complete the statement.
When a and b are nonzero real numbers, and m is an integer, the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

- ☐ a^{b-m} .
☐ $\frac{a^m}{b^m}$.
☐ $a^m - b^m$.



Power of a Quotient

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \text{ and conversely } \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$



Probe students to take notice of the structure of using the laws of exponents. Position students to make connections between the expanded form and the form using the power of a quotient law.



Consider using this routine while students analyze and discuss the table from question 1a. Position students to take note of how the expanded form and the law are connected. Consider challenging students to wonder about their observations and what it means to the other properties of exponents learned previously.



Leverage vocabulary strategies, such as a Frayer Model or foldable, to add the definition and example of the power of a quotient from this page.

2.2.5 Activity: Matching Equivalent Expressions

Component Overview: Students work to determine equivalent monomial expressions. Use questioning if needed to position students to determine the method needed to solve.



2.2.5 Activity: Matching Equivalent Expressions

- Match each expression to an equivalent expression by drawing an arrow connecting them.

Expression	Equivalent Expression
$\frac{1a^4y^4}{256b^4p^4}$	$\frac{a^2}{-4}$
$\left(\frac{8ay}{3bp}\right)^3$	$-2a^3b$
$\frac{-18a^{12}by^4}{9a^9y^4}$	$\frac{100a^6p^7y}{55ab^3p^{13}y}$
$\frac{1}{a^5b^4}$	$\left(\frac{ay}{4bp}\right)^4$
$\frac{20a^5}{11b^3p^6}$	$\frac{a^{20}b^7p^0}{a^{25}b^{11}}$
$\frac{a^2b^3}{-4b^3}$	$\frac{512a^3y^3}{27b^3p^3}$



Consider removing a few of the expressions given in either column, and challenge students to write their own equivalent expressions.



Students benefit from hearing how their peers worked through problems. Consider asking the following.

- How did you determine the matches?
- Did you work the problems in order or was there a method to how you approached the activity?
- Which approach did you use to determine the equivalent expressions (exponent law or expanded form)?

2.2.6 Your Turn!

Component Overview: Students work independently to generate equivalent monomial expressions. Use questioning if needed to position students to determine the method needed to solve.



2.2.6 Your Turn!

- Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

Expression	Equivalent Expression
a. $\frac{s^2t^{18}}{-5st^7}$	$-\frac{st^{11}}{5}$
b. $\left(\frac{0.5d}{10mg}\right)^2$	$\frac{0.0025d^2}{m^2g^2}$ or $\frac{d^2}{400m^2g^2}$
c. $\frac{-204u^2w^8r^{21}}{17u^{16}w^{15}r^4}$	$-\frac{12r^{27}}{u^{14}w^7}$
d. $\left(\frac{e}{k^5}\right)^4$	$\frac{e^4}{k^{20}}$
e. $\frac{49p^4h^9}{81p^{10}}$	$\frac{49h^9}{81p^6}$



Consider giving students choice by positioning them to solve only 4 of the 5 questions.



If students struggle, consider asking the following.

- How does the expression change when a negative exponent is applied?
- Can you write the expression in expanded form to help you generate an equivalent expression?

2.2.7 Wrap-Up: Reflection

Component Overview: Students will assess their own understanding independently and provide feedback regarding the content. Take note of the reasons why the students chose their emoji to possibly address during another lesson.



2.2.7 Wrap-Up: Reflection

- Choose the emoji that reflects your understanding of generating equivalent algebraic expressions to find quotient of two monomials using the properties (laws) of exponents. *Answers will vary.*



0 1 2 3 4 5 6 7 8 9 10

- Complete the statement.

I chose this emoji because *Answers will vary. I understand how use the property, but struggle with decimals. I understand the concept, but I am still struggling when there are a lot of different variables.*



Consider giving students options on how they choose to explain their emoji. Students may be better at explaining their emoji by describing verbally in a quick video or by providing examples of what they understood and did not understand during the lesson, rather than writing out their thoughts.